

Cancer Genomics

Bioinformatic Analysis

Analysis Pipeline Overview



INPUT: FASTQ files (tumor + normal)

OUTPUT: Annotated VCF / MAF

Step 1: Raw Data Quality Control (FastQC)

Key QC Findings

Phred Q30+	Maintained across read length; minor 3' drop typical of Illumina
GC Content	FAIL, deviation from expected distribution, consistent across all 4 samples, likely due to exome enrichment bias
Adapter Content	None detected — trimming not required
Duplication	Within acceptable range for WGS data
N Content	< 0.1% — excellent base calling confidence

QC Outcome per Sample

Normal Sample

PASS

High quality · no adapters · ~50% GC

Tumor Sample

PASS

High quality · no adapters · ~50% GC

No trimming required — both samples proceeded directly to alignment

Step 4: Post-Alignment Quality Control (SAMtools flagstat)

99.80%

Normal
Mapping Rate

99.40%

Normal
Properly Paired

6.65%

Normal
Duplicate Rate

99.18%

Tumor
Mapping Rate

98.78%

Tumor
Properly Paired

28.0%

Tumor
Duplicate Rate

Interpretation

- >99% mapped — excellent alignment
- Proper pairing >98% — insert size consistent
- Normal dup rate 6.7% — good library complexity
- Tumor dup rate 28% — higher PCR amplification (limited input DNA)
- Low singletons — minimal artefacts

Threshold: ≥95% mapping rate required — both samples exceed this benchmark

Step 5–6: Variant Calling & Filtering

Raw Mutect2 Calls: ~23,412

After FilterMutectCalls: 572

97.6% removed

HIGH impact variants: ~45

Cancer driver
genes: 7

Method Details

Mode:	Tumor-normal matched
Reference:	GRCh38 / hg38
Caller:	GATK Mutect2
Filter:	FilterMutectCalls
PoN:	Not used (limitation)
BQSR:	Not applied (limitation)
Output:	Filtered VCF

Key Cancer-Relevant Mutations (VEP Annotation)

Gene	Variant Type	Impact	Biological Role
KMT2C	stop_gained	HIGH	Histone H3K4 methyltransferase (epigenetic TSG)
BCOR	frameshift_variant	HIGH	Polycomb repressor; AML, medulloblastoma
SS18	stop_gained	HIGH	Transcriptional co-activator; sarcoma fusion gene
CIC	frameshift_variant	HIGH	Tumor suppressor; oligodendroglioma, lung
MKI67	frameshift_variant	HIGH	Proliferation marker Ki-67; truncation
ARID1A	inframe_deletion	MODERATE	SWI/SNF chromatin remodelling TSG
ARID1A	missense_variant	MODERATE	SIFT: deleterious · PolyPhen: possibly_damaging

Annotation: Ensembl VEP · HIGH = predicted loss of function · MODERATE = functional change predicted

Biological Interpretation of Findings

Epigenetic Deregulation

- KMT2C — H3K4 methyltransferase; nonsense mutation abolishes SET domain
- Loss disrupts enhancer activity of differentiation genes
- ARID1A — SWI/SNF complex subunit; biallelic inactivation likely
- ARID1A most-mutated TSG in ovarian clear cell & endometrial carcinoma

Transcriptional Repression (BCOR)

- BCOR — BCL6 co-repressor; controls hematopoietic differentiation
- Frameshift → likely nonsense-mediated decay → complete loss
- Associated with AML, medulloblastoma, retinoblastoma
- CIC frameshift further disrupts ETV4/5 transcriptional regulation

Overall Mutational Profile

- Multiple chromatin-remodelling gene co-mutations
- Pattern consistent with endometrial or lymphoma subtypes
- No classical oncogene (KRAS/BRAF/PIK3CA) identified
- Predominantly TSG inactivation — tumour suppressor-driven cancer

Clinical / Therapeutic Implications

- SWI/SNF loss (ARID1A) — potential EZH2 inhibitor sensitivity
- High TMB likely — may predict immunotherapy response
- Epigenetic therapy targeting KMT2C pathway relevant
- Recommend: SV calling, CNV analysis, mutational signatures

Conclusions & Summary

- ✓ Both samples: >99% mapping rate, >98% proper pairing — high-quality WGS data
- ✓ 572 high-confidence somatic variants identified after Mutect2 + FilterMutectCalls
- ✓ Key driver mutations in KMT2C, ARID1A, BCOR, CIC — epigenetic/chromatin pathway
- ✓ Pattern suggests endometrial carcinoma or lymphoma subtype — TSG-driven cancer
- Therapeutic relevance: EZH2 inhibitors, immunotherapy (high TMB likely)
- Future work: SV/CNV calling, BQSR, PoN, mutational signature analysis